

K-FIELD FLYWHEEL FORCE DISTRIBUTION DISK

Full-Unit-Sphere Rotation-Axis Sweep, Exact Continuous-Rim Integration,
Force-Plane Disk Theorem, Closed-Cycle Impulse, and Antipodal
Mass-Distribution Invariance

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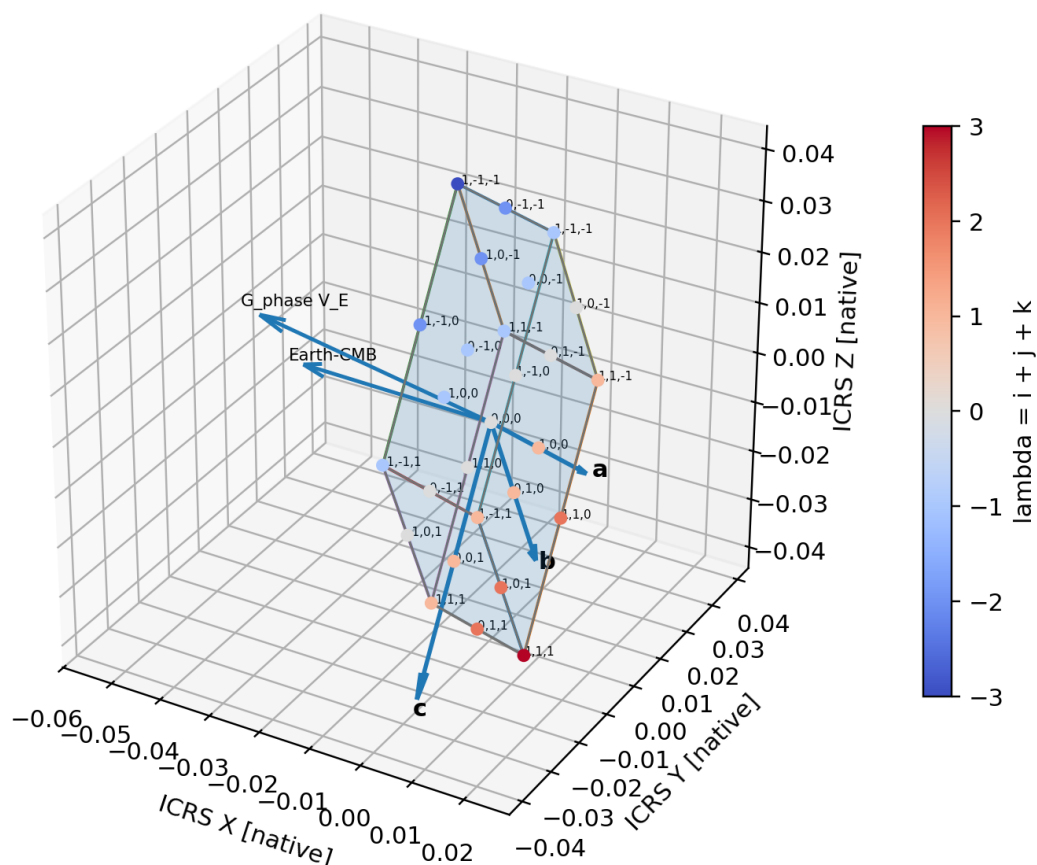
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ICRS Cell3 base cell: 27 centered nodes, six boundary planes, and primitive basis



Title figure. Authoritative ICRS Cell3 base cell in native coordinates. The figure contains all 27 centered half-step nodes, the six boundary face planes, the primitive basis vectors, the Earth-CMB traversal direction, and the transformed vector $\mathbf{G_phase V_E}$. Every coordinate is generated directly from the machine-readable source geometry.

Abstract

This reference develops the complete force geometry of a one-metre-diameter, one-kilogram, uniform zero-thickness flywheel rim whose directed angular-velocity axis is allowed to occupy the full unit sphere in the fixed ICRS K-Field frame. Exact integration over the continuous mass distribution removes the rotational self-term by antipodal odd symmetry and leaves the translation-coupled resultant $F = (\kappa_F M \omega / 2) \mathbf{n} \times (\mathbf{G}_{\text{phase}} \mathbf{V}_E)$. The force is perpendicular to both the rotation axis $\mathbf{n}$ and the transformed Earth-CMB vector $\mathbf{G}_{\text{phase}} \mathbf{V}_E$. Consequently, all possible force vectors lie in one fixed plane, and the image of the complete axis sphere is a filled disk of radius 465412.608635 N per (κ_F rad/s) for $M=1$ kg. The two zero-force axes are plus and minus the unit vector along $\mathbf{G}_{\text{phase}} \mathbf{V}_E$, at ICRS RA 189.285391 deg, Dec 16.626278 deg and its antipode. Maximum force occurs on the great circle of axes perpendicular to $\mathbf{G}_{\text{phase}} \mathbf{V}_E$. The earlier CMB-orthogonal axis sweep is not a separate envelope; it is an ellipse contained within this disk. A uniform solid-angle axis population has mean force vector zero, mean magnitude $(\pi/4)F_{\text{max}}$, RMS magnitude $\sqrt{2/3}F_{\text{max}}$, and an analytic magnitude distribution that is concentrated near the disk boundary. The companion 2:1 dual-axis apparatus is treated separately: its instantaneous force can be transverse, axial, mixed, or zero depending on phase, while the complete cycle impulse is $\mathbf{J} = \pi \kappa_F m \hat{\mathbf{c}} \times (\mathbf{Q} \mathbf{V}_E)$, exactly transverse to the Earth-CMB traversal. For two kilograms of antipodally symmetric rotating mass, this impulse is invariant under redistribution from point masses through finite sectors to a continuous rim. Sector broadening reduces instantaneous phase coherence but cannot increase or decrease the complete cycle impulse.

Field	Authoritative or derived value
Coordinate frame	ICRS; origin at Earth center
Flywheel	Uniform zero-thickness circular rim; diameter 1.0 m; radius 0.5 m; total mass 1.0 kg
Transformed translation vector	$\mathbf{G}_{\text{phase}} \mathbf{V}_E = [-8.802220\text{e}+05, -1.439114\text{e}+05, 2.663350\text{e}+05] \text{ m/s}$
Exact full-sphere force law	$F/(\kappa_F \omega) = (M/2) \mathbf{n} \times (\mathbf{G}_{\text{phase}} \mathbf{V}_E)$
Force locus	Closed disk in the plane perpendicular to $\mathbf{G}_{\text{phase}} \mathbf{V}_E$
Disk radius	4.654126e+05 N per (κ_F rad/s)
Exact zero-force direction	RA 189.285391 deg, Dec 16.626278 deg, and antipode
Sphere sweep	65160 parameter samples; 64442 unique directed axes
Machine-readable companion	K-Field_flywheel_force_distribution_disk_1784810003.json

Contents

Abstract	2
1. Purpose, scope, and model separation	4
2. ICRS coordinate lock and transformed Earth-CMB state	4
3. Authoritative Cell3 base geometry	4
3.1 Boundary plane equations	6
4. Continuous flywheel geometry and mass measure	6
5. Elemental K-Field force law	7
6. Exact continuous-rim integration	8
6.1 Cancellation of the rotational self-term	8

6.2 Surviving translation-coupled term	8
7. Full-unit-sphere axis sweep	8
8. Force-plane disk theorem	9
8.1 Preimages of a force vector	10
9. Exact zero-force axes and maximum-force great circle	11
10. Solid-angle statistics of the disk	11
10.1 Quantiles	13
11. The earlier orthogonal-axis ellipse as a disk cross-section	13
12. Scaling laws and engineering interpretation	14
13. Companion 2:1 dual-axis instantaneous-force theorem	15
14. Closed-cycle impulse and exact transverse closure	16
15. Antipodal mass-distribution invariance	17
16. Numerical validation	18
17. Principal findings	19
18. Consequences and research pathways	19
19. Conclusion	20
Appendix A. Complete 27-node Cell3 registry	21
Appendix B. Cell3 vertices and face-plane ledger	21
B.1 Boundary vertices	21
B.2 Six boundary planes	22
Appendix C. Formula and identity ledger	22
Appendix D. Representative full-sphere sweep ledger	22
Appendix E. Source lock and reconstruction metadata	23

1. Purpose, scope, and model separation

The central objective is to determine the complete net-force vector set generated by the exact continuous-rim law when the directed rotation axis $\mathbf{n}$ is no longer restricted to one great circle. The axis is swept over the complete unit sphere in one-degree polar and azimuthal increments. The calculation is performed in one fixed ICRS frame; neither the Cell3 geometry nor the Earth-CMB vector is rotated between samples. All force values in the primary calculation are normalized as newtons per (κ_F rad/s). A physical force is obtained by multiplying the tabulated value by the source-defined constitutive gain κ_F and the actual angular speed ω in radians per second.

Three related results are presented without conflation. First, the one-kilogram continuous-rim full-sphere model establishes the disk theorem. Second, the one-kilogram continuous-rim great-circle model supplies the earlier ellipse that lies inside the disk. Third, the two-mass 2:1 compound-rotation model supplies the instantaneous phase classes, the closed-cycle impulse theorem, and the antipodal mass-distribution result. Their common geometric operator is the transformed translation vector, but their time paths and total masses are not identical.

The report therefore distinguishes a static orientation map from a time-integrated apparatus. The full-sphere disk answers which instantaneous resultant vectors are geometrically available as the flywheel axis changes. The dual-axis impulse answers what remains after one specified compound rotational cycle. This distinction is essential: an instantaneous force vector, a uniform orientation average, and a closed-path impulse are different mathematical objects.

2. ICRS coordinate lock and transformed Earth-CMB state

The global frame is the International Celestial Reference System with its origin at Earth center. Cartesian vectors are stated as $[x,y,z]$. The Earth-CMB unit traversal vector $\mathbf{c}_{\text{hat}}$ is fixed at $[-0.9707708558, 0.2073480091, -0.1208749295]$, corresponding to RA 167.943299880 deg and Dec -6.942599922 deg. Two unit vectors $\mathbf{e}_0$ and $\mathbf{A}_0$ complete the right-handed orthonormal frame, with $\mathbf{e}_0 \times \mathbf{A}_0 = \mathbf{c}_{\text{hat}}$.

Quantity	ICRS value	RA / Dec or magnitude
$\mathbf{c}_{\text{hat}}$	$[-0.9707708558, 0.2073480091, -0.1208749295]$	RA 167.943300 deg; Dec -6.942600 deg
$\mathbf{e}_0$	$[-0.2008648841, -0.4262314800, 0.8820317590]$	RA 244.767479 deg; Dec 61.888433 deg
$\mathbf{A}_0$	$[0.1313668291, 0.8805302542, 0.4554220324]$	RA 81.514595 deg; Dec 27.092092 deg
$\mathbf{V}_E$	$[-3.590032\text{e}+05, 76679.886034, -44701.060101]$ m/s	3.698125e+05 m/s
$\mathbf{g}=\mathbf{G}_{\text{phase}} \mathbf{V}_E$	$[-8.802220\text{e}+05, -1.439114\text{e}+05, 2.663350\text{e}+05]$ m/s	9.308252e+05 m/s

The symmetric phase metric is

```
G_phase = [[2.9764106378, 1.2431507795, -2.0803600354], [1.2431507795, 2.5922286635,
-2.3178839937], [-2.0803600354, -2.3178839937, 6.7735709133]]
```

Applying this metric to the Earth-CMB translation state gives $\mathbf{g}=\mathbf{G}_{\text{phase}} \mathbf{V}_E$. Its unit direction is $[-0.945636157362, -0.154606293310, 0.286127859460]$. The angle between $\mathbf{c}_{\text{hat}}$ and $\mathbf{g}$ is 31.640861271 deg. This obliquity is the reason the CMB-orthogonal axis sweep produces an ellipse rather than a circle.

3. Authoritative Cell3 base geometry

The force computation is embedded in the same fixed Cell3 geometry used to construct $\mathbf{G}_{\text{phase}}$. The primitive basis vectors $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ are stated in ICRS native coordinates. The base cell is the centered parallelepiped generated by coefficients in $[-1/2, +1/2]$ along each basis vector. The 27-node registry uses all coefficient triples from $\{-1/2, 0, +1/2\}$ cubed. This produces the eight boundary vertices, twelve edge midpoints, six face centers, and the central node.

Basis	ICRS vector [native]	Length [native]
$\mathbf{a}$	$[0.020381438517, -0.003416414496, -0.002332965917]$	0.020797058782
$\mathbf{b}$	$[0.002198745968, 0.012902170259, -0.034066501536]$	0.036494205131

Basis	ICRS vector [native]	Length [native]
c	[-0.000630379485, -0.024257722017, -0.042924354571]	0.049308565900

The primitive volume is 2.842066937810e-05 native^3. Face areas and included angles are:

Face family	Area [native^2]	Included angle	Unit normal in ICRS
A_ab	0.000754721790	83.933483012 deg	[0.1940922008, 0.9131771688, 0.3583792348]
A_ac	0.001011255758	80.448130018 deg	[0.0890525941, 0.8665767693, -0.4910339483]
A_bc	0.001385784432	50.363231814 deg	[-0.9959651903, 0.0836021625, -0.0326192906]

ICRS Cell3 base cell: 27 centered nodes, six boundary planes, and primitive basis

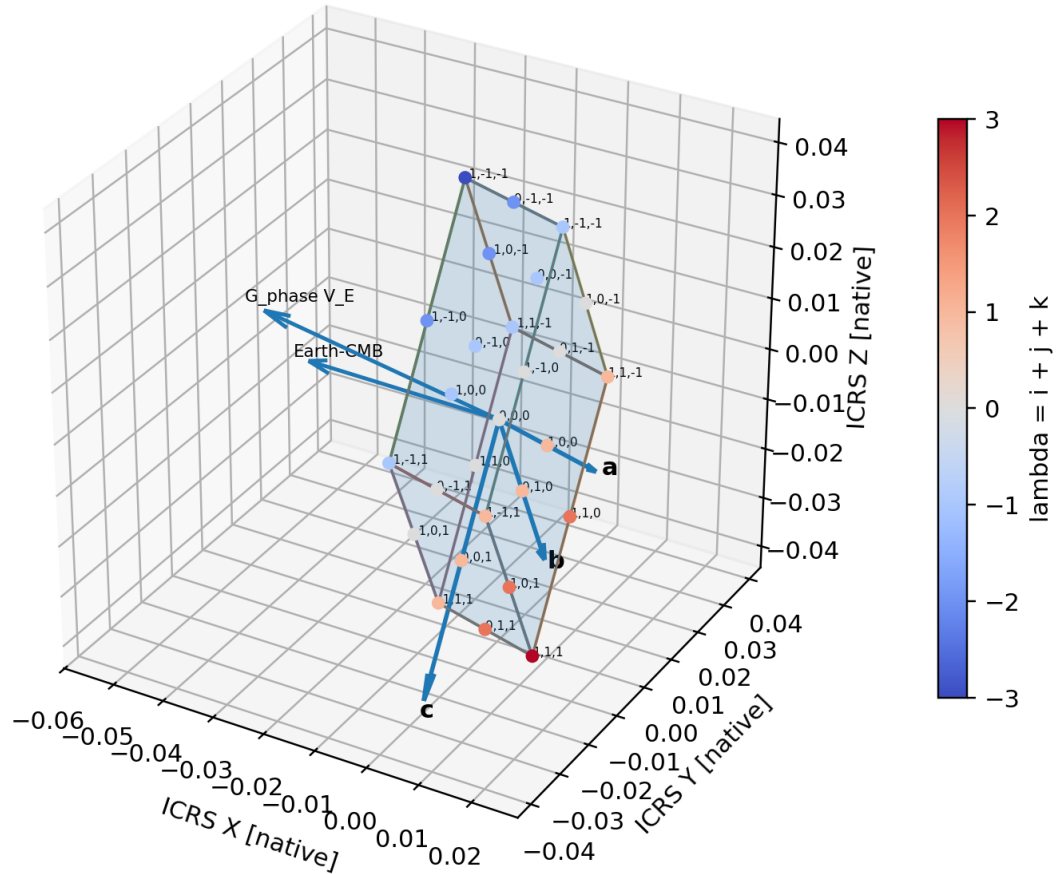


Figure 1. Dimensionally correct Cell3 base cell in the fixed ICRS frame. All 27 nodes are labeled by integer half-step index (i,j,k); the color coordinate is $\lambda = i + j + k$. The six visible boundary surfaces are the paired ab, ac, and bc face planes.

Continuous-rim geometry at the sampled maximum-force axis

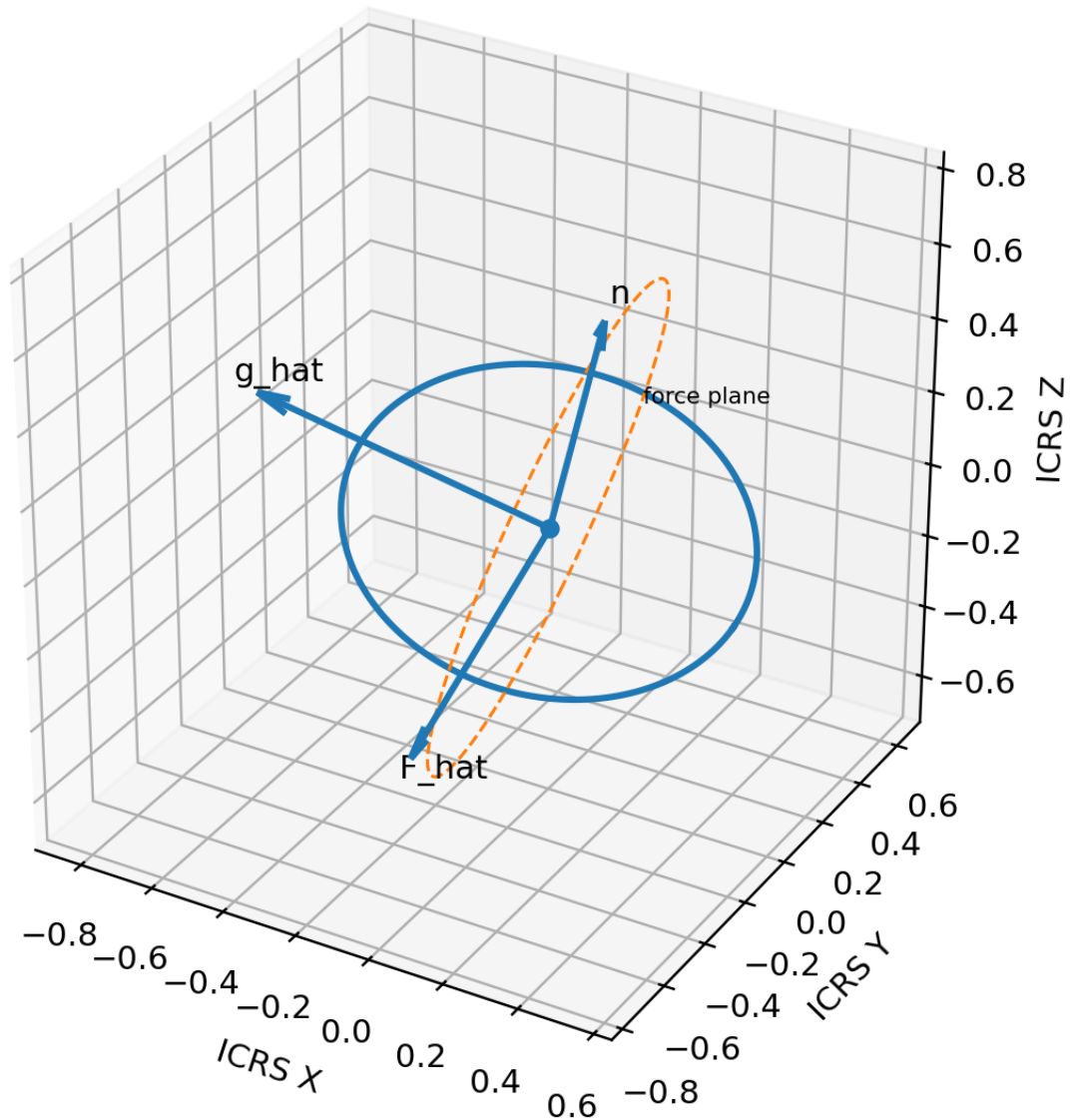


Figure 3. Actual maximum-sample axis geometry. The physical rim radius is 0.5 m. Direction arrows are normalized for display: $\mathbf{n}$ is the sampled rotation axis, $\mathbf{g_hat}$ is the unit transformed Earth-CMB direction, and $\mathbf{F_hat}$ is the corresponding force direction. The dashed circle lies in the force plane perpendicular to $\mathbf{g_hat}$.

5. Elemental K-Field force law

For each rim element, the source force law is radial. Its scalar coefficient contains a translation-rotation cross term and a rotational self-term. With G_{phase} symmetric, the elemental vector is

$$d\mathbf{F} = -\kappa_F dm [\omega \mathbf{t}(\theta)^T G_{\text{phase}} \mathbf{V}_E + R \omega^2 \mathbf{t}(\theta)^T G_{\text{phase}} \mathbf{t}(\theta)] \mathbf{u}(\theta)$$

The first scalar is linear in the Earth-CMB translation and linear in spin rate. The second is quadratic in spin rate and proportional to radius. Both act along the local radial direction $\mathbf{u}$. Therefore every elemental torque about the flywheel center is zero before integration:

$$\frac{d}{dt} \tau = r \times \frac{dF}{dt} = R \times u \times (\text{scalar } u) = 0; \tau_{\text{net}} = 0$$

The absence of net torque does not imply the absence of local structural load. Radial forces still create hoop stress and bearing reactions. It means only that the modeled force does not directly accelerate or brake the rim about its own center.

6. Exact continuous-rim integration

6.1 Cancellation of the rotational self-term

Under the antipodal map $\theta \rightarrow \theta + \pi$, u changes sign while t changes sign. The quadratic scalar $t^T G_{\text{phase}} t$ is unchanged, so the product $[t^T G_{\text{phase}} t]u$ changes sign. Every element therefore has an equal and opposite partner in the integral. The entire $R \omega^2$ self-term cancels exactly for a complete uniform rim.

$$\int_0^{2\pi} [t(\theta)^T G_{\text{phase}} t(\theta)] u(\theta) d\theta = 0$$

6.2 Surviving translation-coupled term

The second moment of a uniform circular rim in its own plane is $(M/2)(I - n n^T)$. Using $t = n \times u$ and the vector identities for a symmetric metric gives a closed-form resultant independent of the arbitrary choice of r_1 and r_2 :

$$\int u u^T dm = (M/2)(I - n n^T)$$

$$F(n) = (\kappa_F M \omega/2) n \times (G_{\text{phase}} V_E)$$

This equation is the foundation of the complete sphere result. It removes the need to integrate thousands of rim elements for every axis orientation. Numerical integration of 16384 elements at independent orientations reproduced the closed form to 3.494884×10^{-9} N per $(\kappa_F \text{ rad/s})$, which is floating-point summation scale relative to force magnitudes near 4×10^5 in the normalized model.

7. Full-unit-sphere axis sweep

The directed axis is parameterized relative to the Earth-CMB frame by polar offset β and azimuth α :

$$n(\beta, \alpha) = \cos(\beta) \hat{c} + \sin(\beta) [\cos(\alpha) \hat{e}_0 + \sin(\alpha) \hat{A}_0]$$

The ledger evaluates $\beta = 0, 1, \dots, 180$ deg and $\alpha = 0, 1, \dots, 359$ deg, producing 65160 parameter rows. At $\beta = 0$ and $\beta = 180$ all 360 azimuth values describe the same physical pole. Removing those repeated pole representations leaves 64442 unique directed axes. The raw ledger preserves every requested step; the analytic results use the unique geometric sphere.

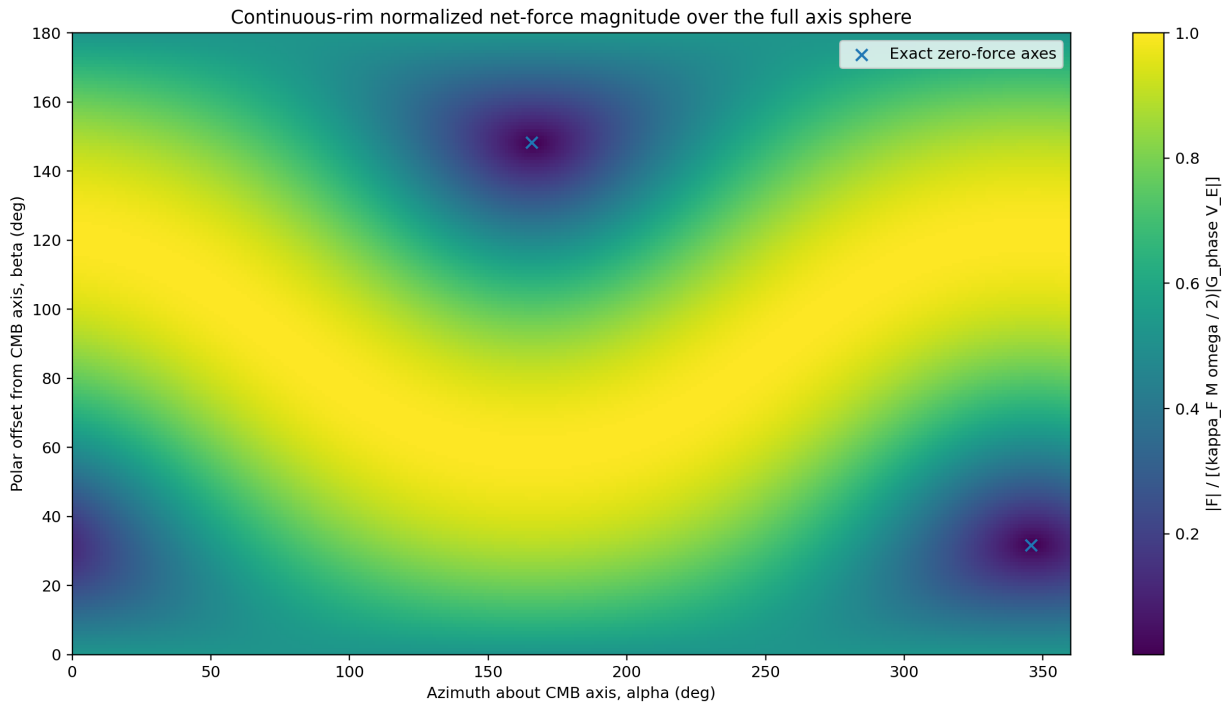


Figure 4. Actual one-degree full-sphere force magnitude map. Color is $|F|$ divided by the analytic maximum. The exact zero-force axes are marked. The bright ridge is the maximum-force great circle where n is perpendicular to $G_{\text{phase}} V_E$.

The computed minimum sample is not exactly zero because a one-degree grid does not land exactly on the analytic null direction. Its force magnitude is 3287.426291 N per (κ_F rad/s), with an angular miss of 0.404710175 deg. The maximum sample differs from the analytic maximum by only $1.809437526 \times 10^{-5}$ N per (κ_F rad/s).

8. Force-plane disk theorem

Let $g = G_{\text{phase}} V_E$ and $C = \kappa_F M \omega / 2$. The force map is the linear skew map $L_g(n) = C n \times g$. It has rank two when g is nonzero. Its kernel is the one-dimensional line parallel to g , and its image is the two-dimensional plane perpendicular to g . Two orthogonality identities follow immediately:

$$F \cdot n = 0; \quad F \cdot g = 0$$

The magnitude depends only on the angle θ between n and g :

$$|F| = C |g| \sin(\theta)$$

As n covers the unit sphere, $\sin(\theta)$ covers every value from zero to one and the force direction covers every azimuth in the plane g perpendicular. Therefore the force endpoint fills the closed disk

$$D_F = \{ f : f \cdot g = 0 \text{ and } |f| \leq C |g| \}$$

For $M=1$ kg and the normalized reference $\kappa_F \omega=1$, the disk radius is $C|g|=465412.608635$ N per (κ_F rad/s). This is not an ellipse and not a shell. It is a filled disk. The boundary circle comes from the great circle of axes perpendicular to g . The center comes from the two axes parallel and antiparallel to g .

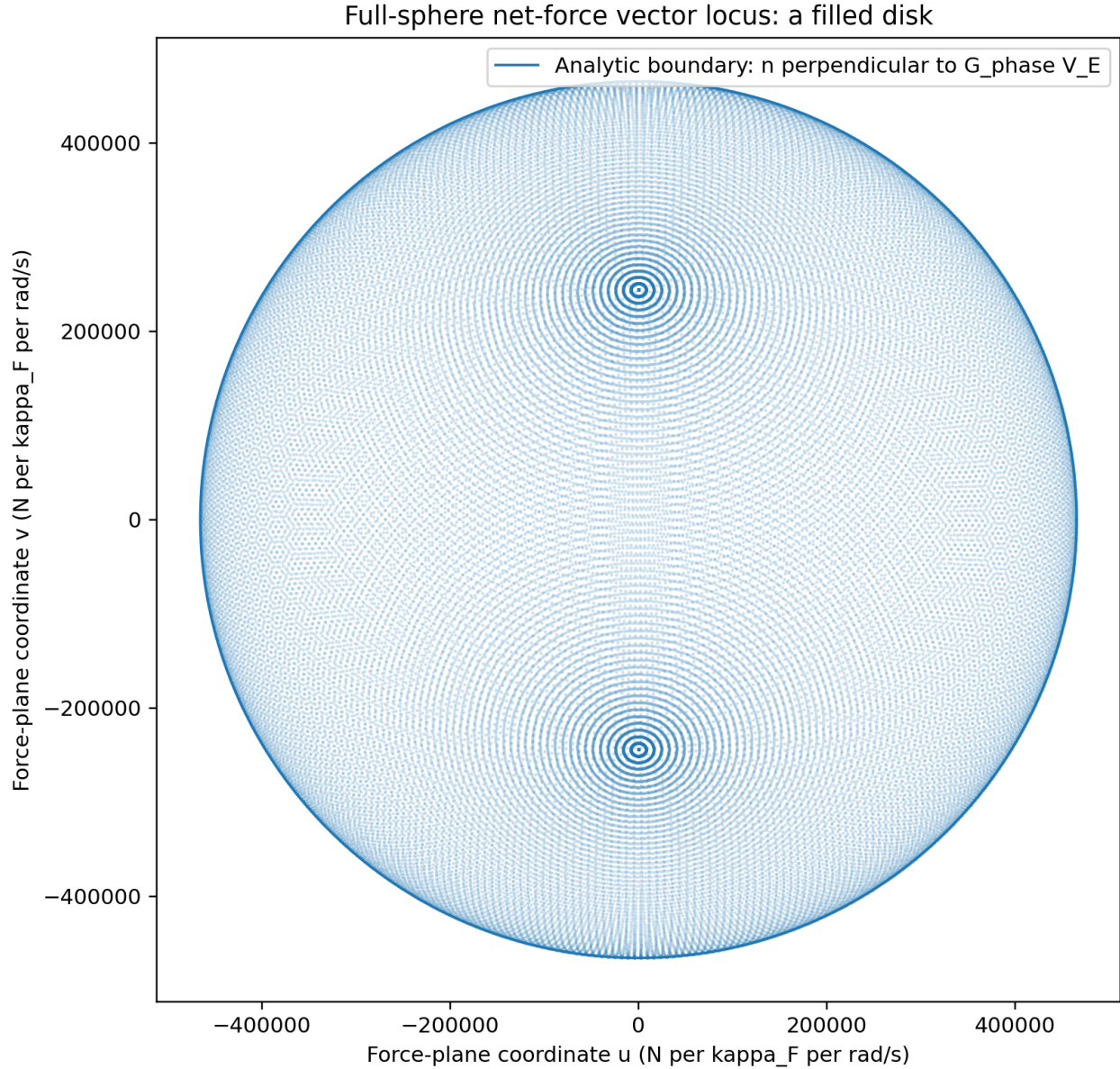


Figure 5. Force vectors from the one-degree full-sphere sweep projected into an orthonormal basis of the force plane. The analytic boundary is a circle of radius 465412.608635 N per (κ_F rad/s). The ring texture is the finite one-degree sampling structure; the analytic continuum is the filled disk.

8.1 Preimages of a force vector

Each interior force vector has two directed-axis solutions. Let $v=f/(C|g|)$. Then $n_{\text{perp}}=g_{\text{hat}} \text{ cross } v$ is fixed, and the component of n along g_{hat} can have either sign:

$$n = g_{\text{hat}} \text{ cross } v \pm \sqrt{1 - |v|^2} g_{\text{hat}}$$

At the disk boundary $|v|=1$, the square-root term vanishes and the directed axis is unique. At the disk center $v=0$, the two preimages are exactly plus and minus g_{hat} . This two-sheet mapping explains why a full sphere collapses to a two-dimensional disk.

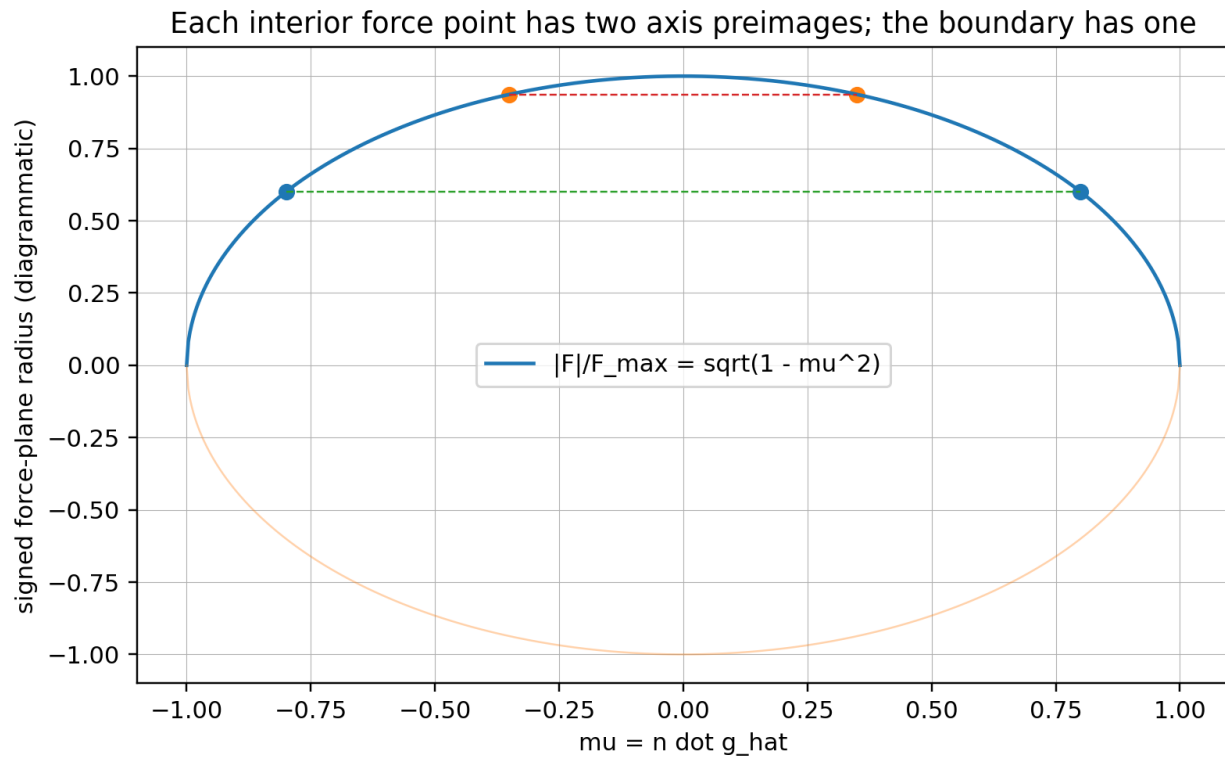


Figure 6. Cross-sectional representation of the two-sheet sphere-to-disk mapping. Interior radii have two values of $\mu = \mathbf{n} \cdot \mathbf{g}_{\text{hat}}$; the boundary has $\mu=0$ and one directed-axis preimage for each boundary force direction.

9. Exact zero-force axes and maximum-force great circle

Force vanishes if and only if $\mathbf{n}$ is parallel or antiparallel to $\mathbf{g}$. The positive null axis is $[-0.945636157362, -0.154606293310, 0.286127859460]$, at RA 189.285391176 deg and Dec 16.626277960 deg. The antipodal null axis is its negative. Relative to the Earth-CMB axis, the positive null is at polar offset 31.640861271 deg and azimuth 345.646138731 deg.

Maximum force occurs when $\mathbf{n} \cdot \mathbf{g} = 0$. This condition defines an entire great circle on the axis sphere. Every axis on that great circle produces a boundary force of the same magnitude, but the force direction changes continuously around the disk boundary. The one-degree grid maximum occurred at $\beta = 118$ deg, $\alpha = 16$ deg and reached 465412.608617 N per (κ_F rad/s).

Extremum	Axis parameters	Axis ICRS	Force ICRS	Magnitude
Exact zero +g	β 31.640861 deg; α 345.646139 deg	$[-0.945636157, -0.154606293, 0.286127859]$	$[0, 0, 0]$	0
Exact zero -g	β 148.359139 deg; α 165.646139 deg	$[0.945636157, 0.154606293, -0.286127859]$	$[0, 0, 0]$	0
Grid minimum	β 148.0 deg; α 166.0 deg	$[0.943381866, 0.156201097, -0.292629240]$	$[-255.435449, 3161.527444, 864.099859]$	3287.426291
Grid maximum	β 118.0 deg; α 16.0 deg	$[0.317237690, -0.244807689, 0.916203822]$	$[33325.672938, -4.454771e+05, -1.305696e+05]$	4.654126e+05

10. Solid-angle statistics of the disk

A uniform distribution of axis directions over the sphere is not uniform over disk area. Let $X = |F|/F_{\max} = \sin(\theta)$. For a uniform axis sphere, $\mu = \cos(\theta)$ is uniform on $[-1, 1]$. Therefore

$$\text{CDF}_X(x) = 1 - \sqrt{1 - x^2}; \quad p_X(x) = x / \sqrt{1 - x^2}, \quad 0 \leq x < 1$$

The probability density rises toward the disk boundary because many sphere directions are nearly perpendicular to g . The exact first and second moments are

$$E[X] = \pi/4; \quad E[X^2] = 2/3; \quad \text{RMS}[X] = \sqrt{2/3}$$

Thus the analytic mean magnitude is 365534.208044 N per (κ_F rad/s), and the analytic RMS magnitude is 380007.803671 N per (κ_F rad/s). The one-degree $\sin(\beta)$ -weighted grid gives 365537.302347 and 380010.636079 respectively. The small differences are the expected finite-grid quadrature error.

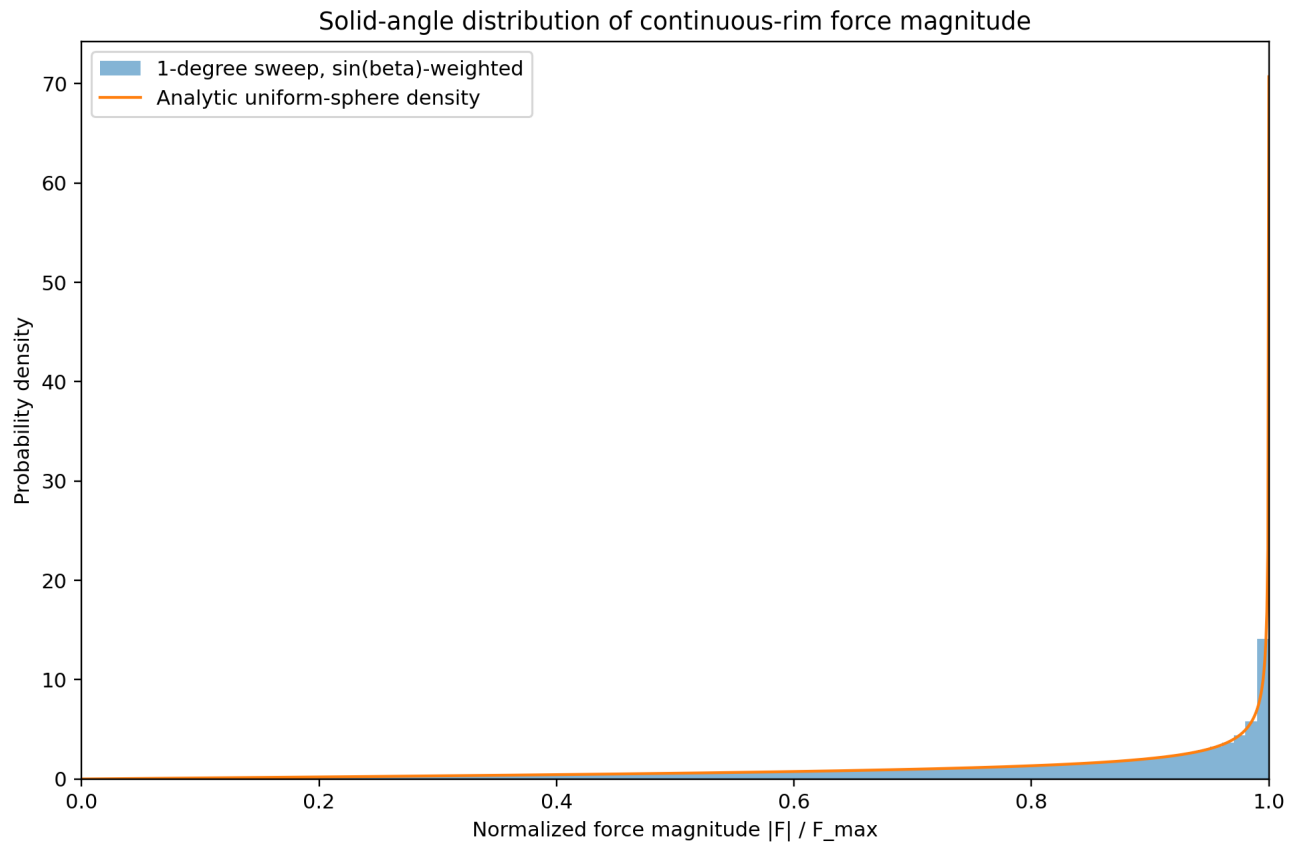


Figure 7. Solid-angle-weighted magnitude distribution from the actual one-degree sweep compared with the exact uniform-sphere density. The integrable density divergence near $F_{\max}$ expresses the concentration of sphere area near axes perpendicular to g .

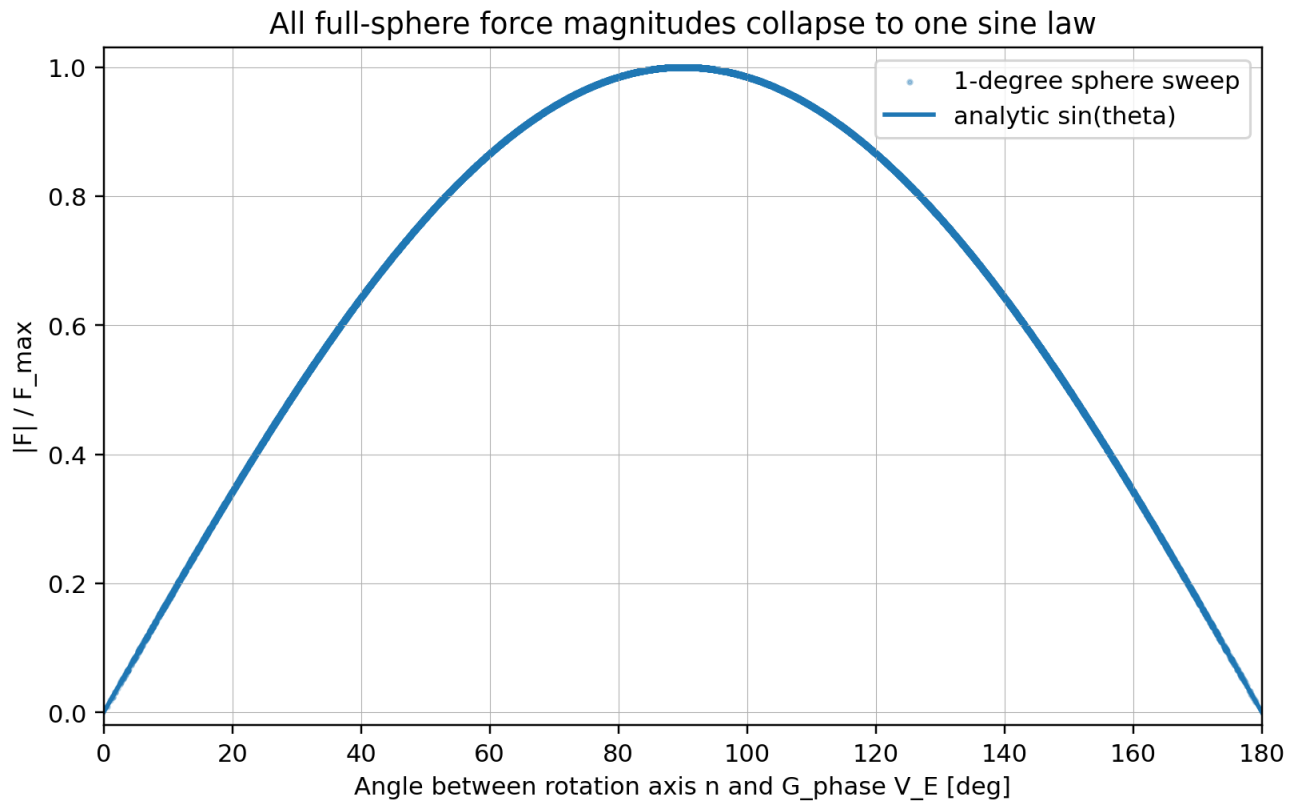


Figure 8. Independent collapse check. Every sampled force magnitude depends only on the angle between n and g and lies on $|F|/F_{\max} = \sin(\theta)$. The visible scatter thickness is smaller than the plotted marker scale.

10.1 Quantiles

The analytic quantile is $x_q = \sqrt{q(2-q)}$. Numeric and analytic values are:

Probability	Numeric force	Analytic force	Numeric - analytic
0.01	65670.846056	65654.527916	16.318140
0.05	1.454219e+05	1.453250e+05	96.821327
0.25	3.078164e+05	3.078415e+05	-25.110552
0.50	4.030625e+05	4.030591e+05	3.376484
0.75	4.506156e+05	4.506338e+05	-18.232913
0.95	4.648374e+05	4.648305e+05	6.898884
0.99	4.653889e+05	4.653893e+05	-0.394812

11. The earlier orthogonal-axis ellipse as a disk cross-section

When the axis is constrained by $n \cdot \hat{c} = 0$, it traces one great circle on the unit sphere. A linear rank-two map sends that great circle to a centered ellipse in the force plane. The ellipse has semi-major radius $C|g| = 465412.608635$ N per (κ_F rad/s) and semi-minor radius $C|g \cdot \hat{c}| = 396230.434489$ N per (κ_F rad/s). The semi-minor to semi-major ratio is $|\cos(31.640861 \text{ deg})|$, the cosine of the obliquity between $\hat{c}$ and g .

$$F(\alpha) / (\kappa_F \omega) = a_F \cos(\alpha) + b_F \sin(\alpha)$$

$$a_F = [6707.042414, -361443.194212, -173135.781945] \text{ N per } (\kappa_F \text{ rad/s})$$

$$b_F = [150028.243957, -217930.035904, 378078.448088] \text{ N per } (\kappa_F \text{ rad/s})$$

The ellipse is therefore not a competing full solution. It is the image of a one-dimensional subset of the axis sphere and lies entirely inside the complete disk. The two ellipse points on the disk boundary occur where the constrained great circle intersects the maximum-force great circle.

The earlier ellipse is a one-great-circle cross-section of the full force disk

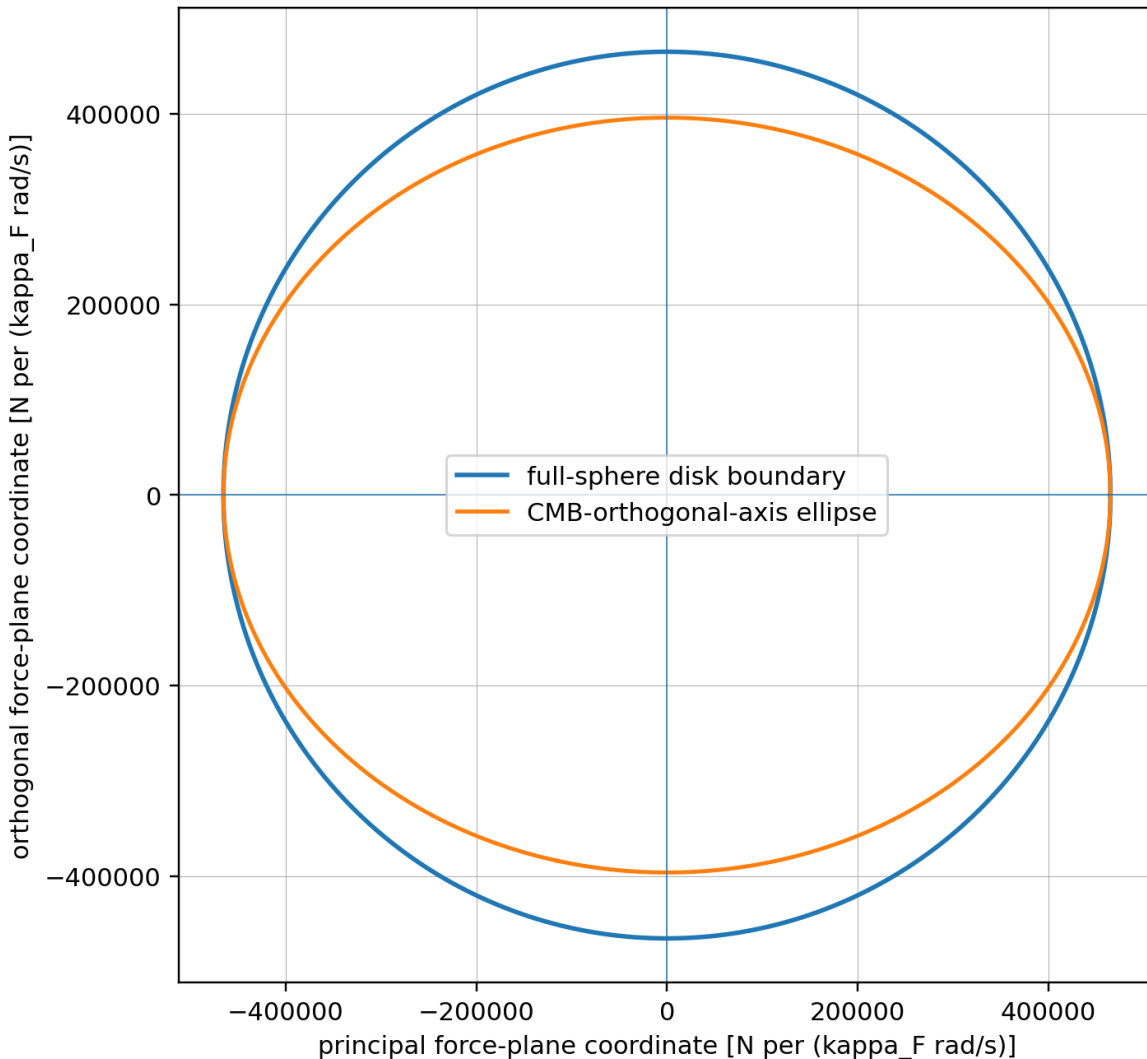


Figure 9. Exact orthogonal-axis ellipse overlaid inside the analytic full-sphere disk. The shared semi-major radius reaches the disk boundary; the semi-minor radius is reduced by the $c_{\hat{t}}$ -to- g obliquity.

12. Scaling laws and engineering interpretation

The exact static-axis resultant is linear in total rim mass, constitutive gain, and spin rate:

$$F = (\kappa_F M \omega/2) \mathbf{n} \text{ cross } \mathbf{g}$$

At fixed angular speed, the net translation-coupled force is independent of rim radius. Radius appears in the canceled self-term but not in the surviving resultant. At fixed rim speed v_{rim} , $\omega = v_{\text{rim}}/R$ and the resultant scales as $1/R$. Reversing spin sends $\mathbf{n}$ to $-\mathbf{n}$ and F to $-F$. A uniform equal-time sweep over an antipodally symmetric set of axes has zero mean force vector, even though the mean magnitude is nonzero.

The disk theorem changes the optimization problem. A three-dimensional search over force direction is unnecessary. The force plane is fixed by $\mathbf{g}$, the maximum magnitude is fixed by $C|\mathbf{g}|$, and orientation control selects a point in that disk.

Static operation at a chosen axis can produce a nonzero normalized force. Uniformly visiting opposite axes produces cancellation. Nonuniform dwell schedules can weight the disk asymmetrically; their result follows directly from the same linear map.

The local radial character also separates center-of-mass response from rotor torque. The model produces a net center-of-mass force with zero torque about the rim center. Mechanical design must therefore resolve radial stress, support reactions, and orientation control separately from spin torque.

13. Companion 2:1 dual-axis instantaneous-force theorem

The instantaneous phase statements belong to the companion dual-axis apparatus containing two one-kilogram endpoint masses at radius 0.5 m. Axis B is aligned with $\hat{c}$ and turns at $\omega_B=1$ rad/s; axis A turns at twice that rate. The endpoint direction is $u(\phi)=\cos(2\phi)\hat{e}(\phi)+\sin(2\phi)\hat{c}$, where $\hat{e}(\phi)=\cos(\phi)\hat{e}_0+\sin(\phi)\hat{A}_0$. The exact system force is

$$F_{\text{net}}(\phi) = -2 \kappa_F m \omega_B [u_{\phi}(\phi)^T Q V_E] u(\phi)$$

The force is radial along $u(\phi)$, so its component parallel to the Earth-CMB axis is

$$F_{\text{parallel_scalar}} = -2 \kappa_F m \omega_B [u_{\phi}^T Q V_E] \sin(2\phi)$$

Four geometric equatorial phases, 0,90,180,270 deg, make u perpendicular to $\hat{c}$ and the force exactly transverse. Four centerline phases, 45,135,225,315 deg, make u parallel or antiparallel to $\hat{c}$ and the force exactly axial. Four additional phases make the scalar $B(\phi)=u_{\phi}^T Q V_E$ vanish, so the complete force is zero. Those roots are 30.402363199, 151.435410918, 230.733949264, and 304.459020416 deg.

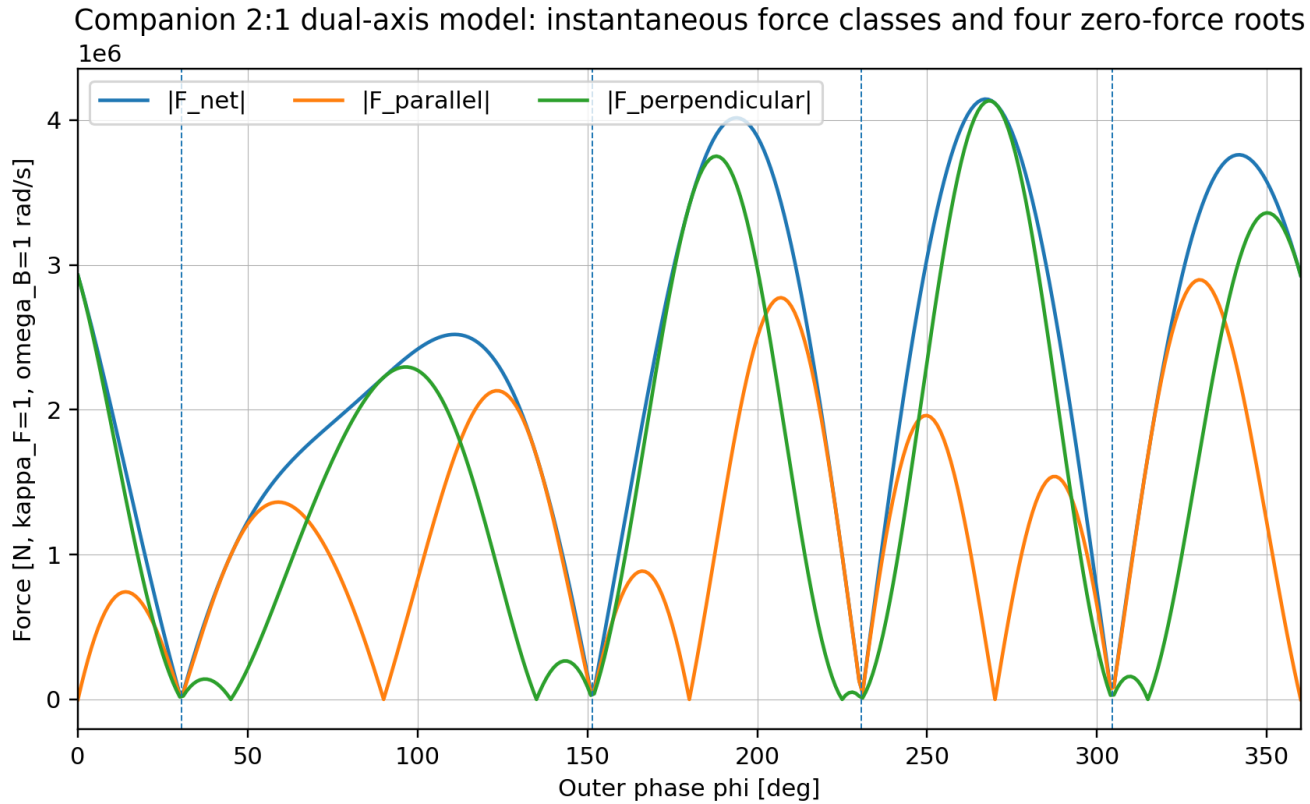


Figure 10. Actual dual-axis one-degree force ledger. The total, axial, and transverse magnitudes vary throughout the cycle. Dashed lines mark the four independently solved Q-metric zero-force phases.

Phase class	Phases [deg]	Condition	Force direction
Transverse	0, 90, 180, 270	$\sin(2\phi)=0$ and $B(\phi)$ not zero	perpendicular to $\hat{c}$
Axial	45, 135, 225, 315	$\cos(2\phi)=0$ and $B(\phi)$ not zero	parallel or antiparallel to $\hat{c}$
Zero force	30.402363, 151.435411, 230.733949, 304.459020	$B(\phi)=0$	undefined because magnitude is zero
Mixed	all remaining phases	both axial and transverse components nonzero	oblique to $\hat{c}$

14. Closed-cycle impulse and exact transverse closure

Instantaneous axial force does not imply an axial closed-cycle residue. Integrating over one complete outer rotation with $dt=d\phi/\omega_B$ gives

$$\mathbf{J}_B = \text{Integral } \mathbf{F}_{\text{net}} dt = \pi \kappa_F m \hat{c} \text{ cross}(\mathbf{Q} \mathbf{V}_E)$$

The cross product makes the closed-cycle impulse exactly perpendicular to $\hat{c}$. For $\kappa_F=1$ and $m=1$ kg per endpoint, the analytic result is

$$\mathbf{J}_B = [118842.532043, 1146514.977540, 1012275.514802] \text{ N s}$$

Its magnitude is $1.534054060\text{e}6$ N s. The rendered dot product with $\hat{c}$ is $1.294\text{e-}11$ N s, corresponding to a relative orthogonality residual of $8.433\text{e-}18$. This nonzero decimal is floating-point residue; the analytic dot product is exactly zero. The one-degree numerical integration differs from the analytic vector by $9.477\text{e-}10$ N s.

During the cycle, the cumulative impulse can contain a large axial component. Its maximum absolute intermediate axial value is $1.964633\text{e}6$ N s. That component returns to zero at cycle closure, while the transverse component terminates at the analytic impulse vector. The cycle-average force equals $\mathbf{J}_B/T_B$ and has magnitude 244152.286620 N at $\omega_B=1$ rad/s.

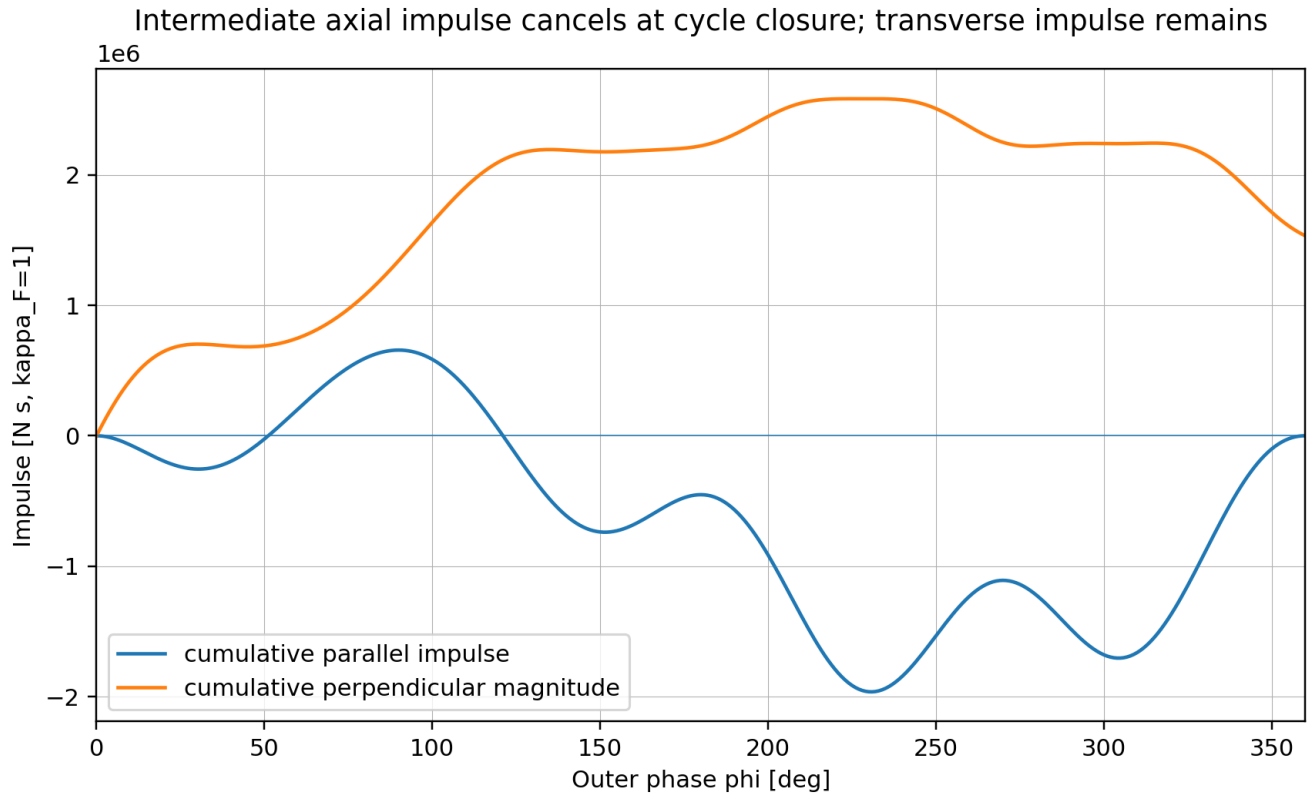


Figure 11. Actual cumulative impulse decomposition. The axial cumulative component grows and reverses but closes at zero. The transverse cumulative magnitude terminates at the exact closed-cycle impulse magnitude.

Quantity	Value
Impulse vector [N s]	[1.188425320e+05, 1.146514978e+06, 1.012275515e+06]
Impulse magnitude [N s]	1.534054060e+06
Impulse RA / Dec	RA 84.082114314 deg; Dec 41.289922334 deg
Dot with $\hat{c}$ [N s]	1.293619088505e-11
Relative orthogonality residual	8.432682538597e-18
One-degree integration error norm [N s]	9.476638023721e-10

15. Antipodal mass-distribution invariance

The dual-axis analysis was extended from two point masses to two diametrically opposite uniform sectors with fixed total mass $M=2$ kg. Sector width $\Delta=0$ is the point-pair limit; $\Delta=\pi$ is a complete uniform rim. The instantaneous sector response decomposes into a distribution-independent rim term plus a distribution-dependent harmonic multiplied by $\text{sinc}(\Delta)=\sin(\Delta)/\Delta$:

$$F_{\Delta} = F_{\text{rim}} + \text{sinc}(\Delta)[F_{\text{point}}(\text{center}) - F_{\text{rim}}]$$

For $0 < \Delta \leq \pi$, $|\text{sinc}(\Delta)|$ decreases monotonically from one to zero. The point-mass limit therefore maximizes the distribution-dependent instantaneous harmonic. Broadening the sectors phase-smears that harmonic. At $\Delta=\pi$, the harmonic is removed completely and only the continuous-rim baseline remains.

The complete one-outer-revolution impulse is independent of Δ and, more generally, independent of every antipodally symmetric angular mass distribution with fixed total mass:

$$J = (\pi \kappa_F M/2) \hat{c} \text{ cross}(Q \mathbf{V}_E)$$

For $M=2$ kg and $\kappa_F=1$ this is the same vector $[118842.532043, 1146514.977540, 1012275.514802]$ N s for the point pair, every finite sector pair, and the full rim. There is therefore no unique pizza-slice angle that increases total impulse. Geometry changes the instantaneous waveform, peak force, and RMS force, but not the cycle impulse.

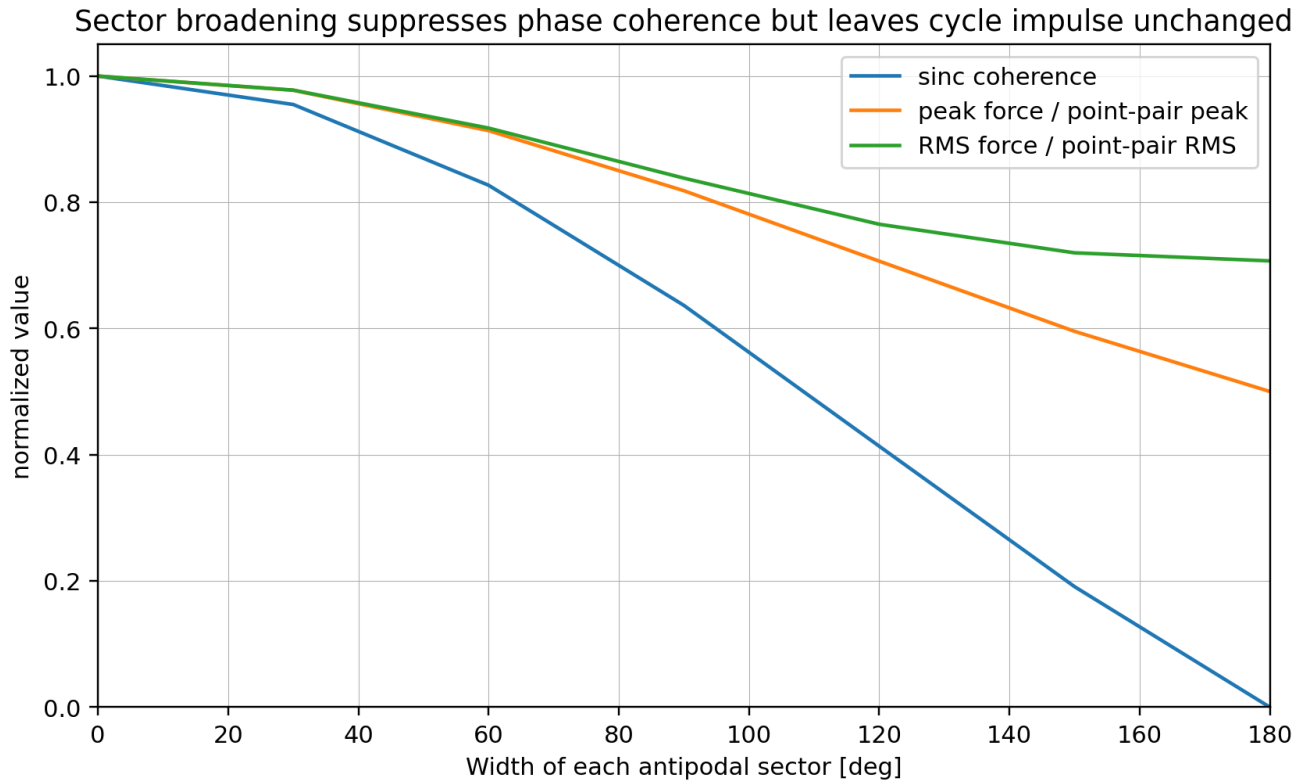


Figure 12. Exact sector-width comparison. Coherence, peak force, and RMS force decrease as each antipodal sector broadens. The cycle impulse is unchanged at every width and is therefore not plotted as a varying curve.

Sector width [deg]	sinc coherence	Peak force [N]	RMS force [N]	Impulse error [N s]
0	1.000000000	4.144924e+06	2.493223e+06	9.386831326e-10
30	0.954929659	4.051518e+06	2.437685e+06	7.986337141e-10
60	0.826993343	3.786377e+06	2.287740e+06	1.397665816e-09
90	0.636619772	3.391844e+06	2.089913e+06	2.445111511e-09
120	0.413496672	2.929454e+06	1.907747e+06	7.594928815e-10
150	0.190985932	2.468353e+06	1.794840e+06	1.098259950e-09
180	3.898171833e-17	2.073226e+06	1.762975e+06	1.173212913e-09

16. Numerical validation

The principal equalities were checked in both closed form and discrete computation. The Cell3 basis reproduces all stated lengths, vertices, and nodes. G_{phase} is exactly symmetric in the stored decimals. The frame $\{e_0, A_0, c_{\text{hat}}\}$ has determinant one and Gram residual $2.220446e-16$. The full-sphere sweep satisfies force-plane and force-axis orthogonality at floating-point scale and closes antipodally.

Validation quantity	Result
G_{phase} symmetry maximum residual	0.000000000000
G_{phase} eigenvalues	1.429328753487, 2.112545026447, 8.800336434665
Sweep-frame Gram residual	2.220446049250e-16
Sweep-frame determinant	1.000000000000
Cell basis-length maximum residual	0.000000000000 native

Validation quantity	Result
Cell volume residual	0.000000000000 native^3
27-node reconstruction maximum residual	0.000000000000 native
Analytic versus 16384-element rim quadrature	3.494884132472e-09 N per (kappa_F rad/s)
Full-sphere max F dot g residual	4.577636718750e-05
Full-sphere max F dot n residual	8.731149137020e-11
Antipodal closure max component	3.492459654808e-10 N per (kappa_F rad/s)
Solid-angle weighted mean vector	[8.391955279338e-12, 9.029313205052e-12, -1.462336035615e-11]

17. Principal findings

The assimilated force model reduces to a compact geometric structure. The exact continuous-rim force is a rank-two linear image of the directed axis. Every possible force vector is confined to the single plane perpendicular to G_{phase} V_E . The complete axis sphere maps onto a filled disk, not an arbitrary three-dimensional cloud. The force is zero only at the two transformed-translation axes and reaches its fixed maximum on the perpendicular great circle.

The earlier great-circle sweep traced a centered ellipse because it sampled only axes perpendicular to the Earth-CMB direction. The ellipse is a literal cross-section of the disk. Its major radius equals the disk radius, while its minor radius is reduced by the cosine of the obliquity between c_{hat} and G_{phase} V_E .

The disk has an exact probability structure under uniform solid angle. Force directions are isotropic within the force plane, the mean vector is zero, the coordinate covariance is isotropic, and magnitudes are concentrated near the boundary. The exact second moment is $E[FF^T] = (F_{\text{max}}^2/3)(I - g_{\text{hat}} g_{\text{hat}}^T)$, so each orthogonal force-plane coordinate has variance $F_{\text{max}}^2/3$ and the total mean-square magnitude is $2F_{\text{max}}^2/3$.

The related dual-axis apparatus demonstrates why instantaneous force and integrated impulse must be distinguished. Axial force can exist at individual phases and substantial axial impulse can accumulate temporarily, yet harmonic closure removes the axial residue over the complete cycle. The final impulse is exactly transverse. Redistributing antipodal mass changes phase coherence but not that impulse.

18. Consequences and research pathways

The force-disk representation supplies a direct control coordinate system. Desired force direction is selected within a fixed plane; desired normalized magnitude determines the required angle between n and g . Every interior force target admits two axis branches, which can be selected according to mechanical clearance, shaft orientation, or continuity of an orientation path. Boundary targets have one branch and require n perpendicular to g .

The map also separates orientation control from constitutive calibration. Geometry determines direction, disk shape, relative magnitude, nulls, extrema, distributions, and scaling. The single multiplicative factor κ_F fixes absolute force normalization. Experiments can therefore test geometry independently through directional ratios and zero locations before absolute gain is established.

The sector theorem identifies two different optimization objectives. Maximizing complete cycle impulse provides no sector-width optimum because the impulse is invariant. Maximizing instantaneous phase coherence selects the point-mass limit. Minimizing the distribution-dependent harmonic selects the continuous rim. Intermediate sectors provide a continuous engineering trade between those endpoints.

The full-sphere result also permits arbitrary orientation-path integration. For any programmed path $n(t)$, the center-of-mass impulse is simply $(\kappa_F M/2) \int \omega(t)[n(t) \times g] dt$. Closed-path cancellation or reinforcement can therefore be evaluated in the fixed two-dimensional force plane rather than by repeated three-dimensional rim integration.

19. Conclusion

The complete continuous-rim axis sweep establishes an exact force-distribution disk. The primary resultant is $F = (\kappa_F M \omega / 2) \mathbf{n} \times (\mathbf{G}_{\text{phase}} \mathbf{V}_E)$. It is perpendicular to the rotation axis and the transformed Earth-CMB vector, and it is independent of rim radius at fixed angular speed. Sweeping $\mathbf{n}$ over the unit sphere fills a disk of radius 465412.608635 N per (κ_F rad/s) in a fixed ICRS plane. The two disk-center axes are the exact null directions; the disk boundary is generated by the maximum-force great circle. The prior ellipse is the image of one constrained great circle within this disk.

The companion dual-axis model adds the time-domain result: instantaneous force alternates among transverse, axial, mixed, and zero states, while the complete cycle impulse closes exactly perpendicular to the Earth-CMB traversal. For fixed total antipodally symmetric mass, that impulse is invariant under redistribution from points to sectors to a continuous rim. The mathematical structure therefore replaces a broad numerical orientation search with explicit planes, nulls, extrema, preimage branches, probability laws, and cycle invariants.

Appendix A. Complete 27-node Cell3 registry

Each node equals $B(i,j,k)/2$ with integer indices i,j,k in $\{-1,0,1\}$. Coordinates are in ICRS native units.

#	i	j	k	lambda	ICRS x	ICRS y	ICRS z
0	-1	-1	-1	-3	-0.010974902500	0.007385983127	0.039661911012
1	-1	-1	0	-2	-0.011290092243	-0.004742877881	0.018199733726
2	-1	-1	1	-1	-0.011605281985	-0.016871738890	-0.003262443559
3	-1	0	-1	-2	-0.009875529516	0.013837068257	0.022628660244
4	-1	0	0	-1	-0.010190719258	0.001708207248	0.001166482958
5	-1	0	1	0	-0.010505909001	-0.010420653760	-0.020295694327
6	-1	1	-1	-1	-0.008776156532	0.020288153386	0.005595409476
7	-1	1	0	0	-0.009091346274	0.008159292378	-0.015866767810
8	-1	1	1	1	-0.009406536017	-0.003969568631	-0.037328945095
9	0	-1	-1	-2	-0.000784183241	0.005677775879	0.038495428054
10	0	-1	0	-1	-0.001099372984	-0.006451085129	0.017033250768
11	0	-1	1	0	-0.001414562727	-0.018579946138	-0.004428926518
12	0	0	-1	-1	0.000315189743	0.012128861009	0.021462177286
13	0	0	0	0	0.000000000000	0.000000000000	0.000000000000
14	0	0	1	1	-0.000315189743	-0.012128861009	-0.021462177286
15	0	1	-1	0	0.001414562727	0.018579946138	0.004428926518
16	0	1	0	1	0.001099372984	0.006451085129	-0.017033250768
17	0	1	1	2	0.000784183241	-0.005677775879	-0.038495428054
18	1	-1	-1	-1	0.009406536017	0.003969568631	0.037328945095
19	1	-1	0	0	0.009091346274	-0.008159292378	0.015866767810
20	1	-1	1	1	0.008776156532	-0.020288153386	-0.005595409476
21	1	0	-1	0	0.010505909001	0.010420653760	0.020295694327
22	1	0	0	1	0.010190719258	-0.001708207248	-0.001166482958
23	1	0	1	2	0.009875529516	-0.013837068257	-0.022628660244
24	1	1	-1	1	0.011605281985	0.016871738890	0.003262443559
25	1	1	0	2	0.011290092243	0.004742877881	-0.018199733726
26	1	1	1	3	0.010974902500	-0.007385983127	-0.039661911012

Appendix B. Cell3 vertices and face-plane ledger

B.1 Boundary vertices

#	a coeff	b coeff	c coeff	ICRS x	ICRS y	ICRS z
0	-0.5	-0.5	-0.5	-0.010974902500	0.007385983127	0.039661911012
1	-0.5	-0.5	0.5	-0.011605281985	-0.016871738890	-0.003262443559
2	-0.5	0.5	-0.5	-0.008776156532	0.020288153386	0.005595409476
3	-0.5	0.5	0.5	-0.009406536017	-0.003969568631	-0.037328945095
4	0.5	-0.5	-0.5	0.009406536017	0.003969568631	0.037328945095
5	0.5	-0.5	0.5	0.008776156532	-0.020288153386	-0.005595409476
6	0.5	0.5	-0.5	0.011605281985	0.016871738890	0.003262443559
7	0.5	0.5	0.5	0.010974902500	-0.007385983127	-0.039661911012

B.2 Six boundary planes

Family	Plane	n_x	n_y	n_z	d
ab faces (constant c coefficient)	1	0.194092200808	0.913177168821	0.358379234791	-0.018828573500
ab faces (constant c coefficient)	2	0.194092200808	0.913177168821	0.358379234791	0.018828573500
ac faces (constant b coefficient)	1	0.089052594102	0.866576769291	-0.491033948326	0.014052166903
ac faces (constant b coefficient)	2	0.089052594102	0.866576769291	-0.491033948326	-0.014052166903
bc faces (constant a coefficient)	1	-0.995965190309	0.083602162507	-0.032619290564	-0.010254361619
bc faces (constant a coefficient)	2	-0.995965190309	0.083602162507	-0.032619290564	0.010254361619

Appendix C. Formula and identity ledger

Construct	Exact expression
Axis sphere	$n(\beta, \alpha) = \cos(\beta) \hat{c} + \sin(\beta) [\cos(\alpha) \hat{e}_0 + \sin(\alpha) \hat{A}_0]$
Rim point	$r = R u(\theta), u =1, u \cdot n = 0$
Rim tangent	$t = n \times u$
Local velocity	$v = V_E + R \omega t$
Elemental force	$dF = -\kappa_F dm [\omega t^T G_{\text{phase}} V_E + R \omega^2 t^T G_{\text{phase}} t] u$
Self-term cancellation	$\text{Integral} [t^T G_{\text{phase}} t] u d\theta = 0$
Rim second moment	$\text{Integral} u u^T dm = (M/2)(I - n n^T)$
Continuous-rim resultant	$F = (\kappa_F M \omega/2) n \times (G_{\text{phase}} V_E)$
Force orthogonality	$F \cdot n = 0$ and $F \cdot (G_{\text{phase}} V_E) = 0$
Force magnitude	$ F = (\kappa_F M \omega/2) G_{\text{phase}} V_E \sin(\theta)$
Disk	$f \cdot g = 0$ and $ f \leq F_{\text{max}}$
Disk preimages	$n = \hat{g} \times (f/F_{\text{max}}) \pm \sqrt{1 - f ^2/F_{\text{max}}^2} \hat{g}$
Uniform-sphere CDF	$P(F /F_{\text{max}} \leq x) = 1 - \sqrt{1 - x^2}$
Uniform-sphere density	$p(x) = x/\sqrt{1 - x^2}$
Mean magnitude	$E F = (\pi/4) F_{\text{max}}$
RMS magnitude	$\sqrt{E F ^2} = \sqrt{2/3} F_{\text{max}}$
Orthogonal-axis ellipse	$F/(\kappa_F \omega) = a_F \cos(\alpha) + b_F \sin(\alpha)$
Dual-axis force	$F_{\text{net}} = -2 \kappa_F m \omega_B [u \cdot \hat{\phi}]^T Q V_E u$
Dual-axis cycle impulse	$J_B = \pi \kappa_F m \hat{c} \times (Q V_E)$
Sector interpolation	$F_{\text{Delta}} = F_{\text{rim}} + \text{sinc}(\Delta) [F_{\text{point}} - F_{\text{rim}}]$
Antipodal impulse	$J = (\pi \kappa_F M/2) \hat{c} \times (Q V_E)$

Appendix D. Representative full-sphere sweep ledger

The complete 65160-row one-degree ledger is embedded in the machine-readable companion. The following rows expose the coordinate convention and representative magnitudes.

beta	alpha	axis RA	axis Dec	F_x	F_y	F_z	F
0	0	167.943300	-6.942600	18914.376424	1.824735e+05	1.611087e+05	2.441523e+05
30	0	182.041443	19.653746	19733.851687	-22694.888068	52956.294387	60900.344515
30	90	141.348725	7.067020	91394.452459	49061.691086	3.285634e+05	3.445488e+05
60	0	201.929955	44.702398	15265.657327	-2.217822e+05	-69385.659493	2.328836e+05
60	90	113.219520	19.509891	1.393855e+05	-97496.184338	4.079799e+05	4.420197e+05
90	0	244.767479	61.888433	6707.042414	-3.614432e+05	-1.731358e+05	4.008269e+05
90	90	81.514595	27.092092	1.500282e+05	-2.179300e+05	3.780784e+05	4.614600e+05
90	180	64.767479	-61.888433	-6707.042414	3.614432e+05	1.731358e+05	4.008269e+05
90	270	261.514595	-27.092092	-1.500282e+05	2.179300e+05	-3.780784e+05	4.614600e+05
120	16	324.609310	64.611581	31939.355601	-4.441518e+05	-1.344352e+05	4.651492e+05
148	166	9.401492	-17.015431	-255.435449	3161.527444	864.099859	3287.426291
180	0	347.943300	6.942600	-18914.376424	-1.824735e+05	-1.611087e+05	2.441523e+05

Appendix E. Source lock and reconstruction metadata

Source artifact	Bytes	SHA-256
K-Field_continuous_rim_full_sphere_axis_sweep_analysis_1784808099 (1).json	5423	fb10064ae5e13d4cba28d1efe8dfff40660031fc42fbc916ba5d9e02347046ab
K-Field_flywheel_force_analysis_1784808116.json	2316139	a965becf29845009e05ca992befb7699a5869415b5296c315bdf81fef10fa4e8
K-Field_dual_axis_kinetic_mass_CMB_traversal_net_force_vector_1784751310.json	586773	dbd84eae04dcf4d31eb3fa86a44571a9676cb6ddb852b900204fca97aca0f16b
K-Field_rim_sector_optimization_summary.json	6695	e47246fe8685f97fe870cfafcf43e06f91c1d93827313f471136ab0d198ed42
K-Field_continuous_rim_full_sphere_axis_sweep_1deg_1784808099.csv	17635004	f95dae2d358b6cb4ea8c2c41d6e96b988c012eb64bca17804ee1e7b47b81ae32
01_full_sphere_force_magnitude_map.png	218573	e7219c4def3e6f38e818189ed557afa09f50499f24733f053e64fc5abc7b4d8d
02_full_sphere_force_vector_disk.png	1258293	700bc3834ed39b2903865de42a062d3ed8f89ab6ba0bb260983f6bb97226f58c
03_solid_angle_force_magnitude_distribution.png	90113	17b9c6d15a857724bb4688d4a38056d034dbe7f49e1aa62427a0124b7bc7ae64

Generated Unix epoch: 1784810003. The PDF and JSON companion are generated from the same loaded source objects. Figures 1-3 and 6, 8-12 are regenerated from those numerical objects; Figures 4, 5, and 7 are the authoritative full-sphere outputs stored with the sweep analysis.